Fine-Tuning Cross-Encoders for Detecting Hallucinations in Financial Customer Service Chatbots

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Abstract

Mitigating hallucinations in Large Language Model (LLM) systems remains a key barrier to deployment in higher-risk domains such as finance and medicine where factual accuracy is both a regulatory requirement and component of responsible AI practice. This paper finetunes a cross-encoder DeBERTa-v3 transformer as a lightweight hallucination detection model to identify generated responses that are unsupported by or contradict retrieved evidence. The model is trained on Halubench benchmark data and a synthetic, domain-specific dataset simulating retail bank customer service chatbot responses. Its performance is evaluated against two baselines: a zero-shot prompted LLM-as-a-judge, based on the Lynx framework from Ravi et al. (2024), and a commercial AWS Bedrock contextual grounding guardrail. Results show sufficient performance for the cross-encoder model to be considered as a scalable, low latency component of hallucination detection in production-grade systems.

1 Introduction

Large Language Models (LLMs) are increasingly deployed in customer-facing systems, supporting tasks from dialogue generation to decision assistance (Brown et al., 2020; OpenAI, 2023). Despite their fluency and versatility, LLMs frequently produce hallucinations—outputs that are unverifiable or contradict retrieved evidence (Ji et al., 2023). In high-risk domains such as finance, these errors can erode user trust, and introduce compliance and reputational risks.

Retrieval-Augmented Generation (RAG) is a commonly used technique for mitigating hallucinations by incorporating externally retrieved documents into the model prompt to support response grounding (Lewis et al., 2020). However, hallucinations persist even in RAG settings, often due to incomplete retrieval, poor integration of evidence, or continued reliance on a model's intrinsic knowledge. As a result, post-generation hallucination detection has become an important component of production-grade LLM systems, functioning as a safeguard or 'guardrail' before output delivery.

This paper aims to fine-tune a lightweight cross-encoder on benchmark and synthetic financial data for hallucination detection. It compares this model against an open-source LLM-as-a-judge and a commercial grounding guardrail to assess its viability as a low-latency, scalable guardrail in production LLM-based systems for higher-risk domains like finance.

Hallucination detection is typically framed as a *groundedness classification* task: for a given query, retrieved context, and model-generated response triplet, the guardrail must determine whether the response is supported by the evidence provided (Luo et al., 2024). Effective solutions must handle semantic nuance, domain-specific language, and variable response structures.

1.1 Background

Several groundedness evaluation frameworks have emerged to address this challenge, with LLM-based approaches among the most widely adopted (Deng et al., 2023; Luo et al., 2024). **RAGAS** (Es et al., 2023), an open-source toolkit based on heuristic metrics, evaluates answer faithfulness by prompting an LLM to extract factual claims from a response and verify each claim against the retrieved context, returning a score based on the proportion of grounded claims.

LLM-as-a-judge methods, including both prompted and fine-tuned models, assess hallucinations by reasoning over the full query, context, and response to provide a more holistic assessment of groundedness. Prompted approaches use general-purpose models such as GPT-4 to classify responses relative to retrieved context (Zheng et al., 2023; Manakul et al., 2023), while fine-tuned LLM-as-a-judges such as the open-source **Lynx** (Ravi et al., 2024) model are explicitly trained on hallucination datasets. Lynx has demonstrated stronger performance than GPT-4 on QA-style hallucination detection tasks.

Some commercial guardrail systems—e.g., AWS Bedrock's contextual grounding guardrail—offer API-based hallucination checks for LLM outputs, but their inner mechanisms are opaque and they may introduce latency, cost, and privacy risks.

While performant, LLM-based evaluators present several practical limitations: they are costly to run at scale, introduce latency, and produce non-deterministic outputs—where repeated evaluations of the same input can yield different results. This limits reproducibility, complicates auditing, and creates inconsistencies in quality assurance workflows. These challenges motivate the need for efficient, locally deployable alternatives.

Encoder-based models offer a competitive alternative to LLM-based hallucination detection, combining transparency, low inference overhead, and real-time feasibility. **Luna** is a DeBERTa-large cross-encoder fine-tuned on multi-domain RAG data to detect hallucinated spans at the token level (Belyi et al., 2024). It outperforms GPT-3.5-based evaluators on the RAGTruth benchmark while achieving up to 97% cost reduction and 91% lower latency. **Lettuce Detect**, built on the ModernBERT architecture, surpasses Luna and GPT-4-turbo with a 79.2% F1—despite being 30× smaller than top LLMs. (Ádám Kovács and Recski, 2025).

1.2 Contributions

This paper builds on recent work by addressing key limitations of LLM-based hallucination detection, and proposes a lightweight cross-encoder classifier based on DeBERTa-v3 for groundedness classification. The model is fine-tuned on a combination of benchmark data and a synthetic financial-domain dataset simulating retail bank Q&A customer service interactions Its performance is evaluated against two LLM-based baselines: an open-source zero-shot verifier based on the Lynx framework Ravi et al. (2024), and a closed-source commercial grounding guardrail provided by AWS Bedrock. The contributions of this paper are:

1. Develop a practical, resource-efficient cross-encoder approach for hallucination detection in RAG systems that balances performance and latency;

2. Construct a domain-specific evaluation set reflecting the language and groundedness challenges of retail bank dialogue systems, leveraging the Banking77 dataset to enrich task realism and intent coverage[1];

3. Conduct a comparative evaluation of the cross-encoder against both open-source and commercial LLM-based evaluators, highlighting trade-offs in groundedness accuracy, transparency, and deployability.

Together, these findings offer actionable guidance for operationalising hallucination detection in higher-risk environments such as banking and finance, where LLM outputs must be both reliable and auditable.

2 Methodology

2.1 Data Construction

The training and evaluation data were sourced from two key datasets: (i) the HaluBench benchmark dataset; and (ii) a synthetic dataset of customer service queries for a retail-banking chatbot.

HaluBench Benchmark

This benchmark dataset contains around 15,000 query–context–response triplets annotated for hallucination detection. Introduced by Ravi et al. (2024) to develop the LYNX model, HaluBench combines multiple established datasets from different domains.[2] It includes both naturally occurring hallucinations generated by large language models and synthetic perturbations. The benchmark consists of the following domain-specific subsets:

- **FinanceBench**—financial reasoning (Islam et al., 2023)
- **DROP**—numerical reasoning (Dua et al., 2019)
- **PubMedQA**—biomedical question answering (Jin et al., 2019)
- **RAGTruth**—retrieval-augmented generation evaluation (Wu et al., 2023)

Two sub-datasets-COVID-QA (Möller et al., 2020) and HaluEval (Li et al., 2023) datasets- were

[1] https://huggingface.co/datasets/PolyAI/banking77

[2] See Section 3.2 of Ravi et al. (2024) for further detail on the construction of the HaluBench benchmark dataset.

excluded from analysis due to lengthy report-style contexts and limited domain relevance, and due to a systematic length bias between grounded and hallucinated responses, respectively.

A final set of 3,900 observations were sampled from Halubench for both fine-tuning and evaluation (with 3,315 for training and 585 for the hold-out evaluation set).[3]

RetailbankQA Synthetic Dataset

To fine-tune a hallucination detection model in the context of a retail bank customer service setting, a synthetic RetailbankQA dataset was constructed using a multi-stage prompting pipeline shown in Figure 1.

- **Step 1: Query Sampling.** A representative sample of 1,385 customer queries were obtained from the Banking77 dataset, covering 77 intent categories (e.g., `card_payment_problems`,`atm_support`, `interest_rate_query`).[4]
- **Step 2: Context Generation.** For each query from Step 1, `gpt-4o-mini` was prompted to generate a 150–250 word FAQ-style passage that could plausibly be used to answer the customer's question in a banking customer service setting.
- **Step 3: Response Generation.** Two responses were created for each query-context pair:
  - **Grounded responses** were generated using `gpt-4o` with temperature set to zero, and prompted to rely strictly on the accompanying context passage, without introducing any external knowledge.
  - **Hallucinated responses** were generated using `gpt-3.5-turbo-0125`, with prompting designed to subtly introduce factual inconsistencies (e.g., fabricated service features or incorrect fees) while maintaining surface fluency and plausible relevance.
- **Step 4: Dataset Assembly.** The final dataset included 2,770 labeled (query, context, response) triplets—1,385 grounded and 1,385 hallucinated. Of these, 1,000 grounded–hallucinated pairs were allocated for training, and 385 pairs were held out as an evaluation set. The train–validation split was performed using unique pair identifiers to ensure no query–context–response overlap across splits, thereby preventing data leakage. All examples were formatted consistently with the HaluBench schema. An illustrative example test case is provided in Appendix A (see Table 3).
- **Step 5: LLM-based Annotation Verification.** To assess label quality, a random sample of 50 examples (25 grounded, 25 hallucinated) from the synthetic dataset was reviewed by OpenAI's `o3-pro` model[5], prompted to act as an expert human QA reviewer and annotator. This model offers stronger reasoning capabilities than those used during data construction (`gpt-3.5-turbo-0125` for hallucinations; `gpt-4o-mini` for context generation; and `gpt-4o` for grounded responses). Following strict groundedness criteria, the annotation process showed a 90% agreement with the original labels. This degree of alignment suggests that the synthetic labels are broadly consistent with expert judgment and sufficiently reliable for training purposes.[6]

Final sample

The final sample consists of 6,600 observations, with 3,900 from HaluBench and 2,770 test cases from the constructed RetailbankQA dataset.

Domain	# Examples	%H	Med CT	Max CT
RetailBankQA	2,770	50%	275	352
HaluBench	3,900	34%	291	2494
DROP	1,000	50%	224	1204
FinanceBench	1,000	50%	454	2494
PubMedQA	1,000	50%	280	937
RAGTruth	900	18%	282.5	538

Table 1: Summary statistics of full training and evaluation dataset, where **%H** denotes share of hallucinated responses and **CT** indicates context token length. HaluBench includes benchmark datasets from finance, biomedical, and general knowledge domains.

[3]The reduction in sample size compared to the full HALUBENCH dataset reflects the exclusion of the HALUEVAL subset (approximately 10,000 examples), which was removed due to a systematic length bias between hallucinated and grounded responses.

[4]The Banking77 dataset contains 13,083 real-world customer service queries from the banking domain. It is widely used for research in intent classification and banking-domain conversational AI.

[5]https://platform.openai.com/docs/models/o3

[6]Disagreements primarily involved borderline cases with minor omissions or over-specificity, underscoring the need for clearer prompting or light label smoothing in future iterations.

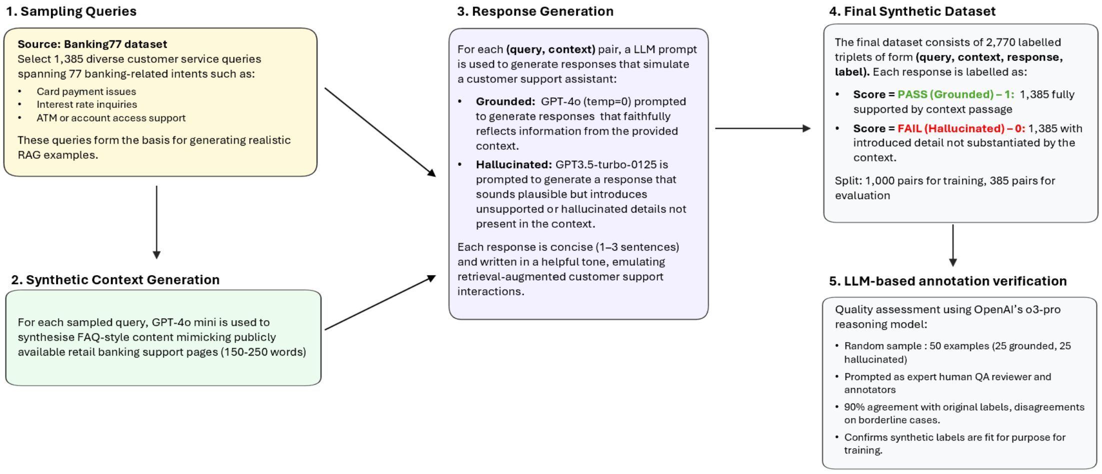

Figure 1: Synthetic data construction pipeline showing the four stages: sampling customer queries, generating synthetic contexts, creating grounded and hallucinated responses, and producing the final labeled dataset.

Table 1 reports token-level statistics across all datasets. The synthetic retail bank dataset (RetailbankQA) contains moderate context lengths compared to other datasets in HaluBench. The share of hallucination are balanced in all datasets but RAGTruth. Each dataset was split using stratified sampling into training, validation, and test subsets (70/15/15) to preserve class balance. Validation sets were used for hyperparameter tuning, and final evaluation was conducted on held-out test sets. All examples were tokenized using the DeBERTa-v3-base tokenizer.

Dataset	Model	Accuracy	Macro Precision	Macro Recall	Macro F1
HaluBench	DeBERTa-v3 Cross-encoder	69.0%	70.0%	68.0%	68.0%
	GPT-4o (LLM Evaluator)	82.6%	82.9%	83.6%	82.5%
	AWS Bedrock	58.6%	60.5%	60.3%	58.6%
- DROP	DeBERTa-v3 Cross-encoder	88.0%	85.9%	89.7%	87.8%
	GPT-4o (LLM Evaluator)	82.4%	82.7%	82.2%	82.2%
	AWS Bedrock	47.9%	46.7%	47.1%	45.8%
- FinanceBench	DeBERTa-v3 Cross-encoder	52.2%	51.9%	75.6%	61.5%
	GPT-4o (LLM Evaluator)	84.8%	86.1%	84.9%	84.7%
	AWS Bedrock	50.6%	51.1%	50.8%	47.3%
- RAGTruth	DeBERTa-v3 Cross-encoder	75.8%	85.1%	85.1%	85.1%
	GPT-4o (LLM Evaluator)	84.8%	74.8%	77.2%	75.9%
	ASW Bedrock	80.8%	69.9%	74.8%	71.7%
- pubmedQA	DeBERTa-v3 Cross-encoder	68.6%	67.0%	82.9%	74.1%
	GPT-4o (LLM Evaluator)	77.9%	82.0%	79.4%	77.6%
	AWS Bedrock	60.0%	62.2%	61.3%	59.6%
RetailbankQA	DeBERTa-v3 Cross-encoder	97.0%	97.0%	97.0%	97.0%
	GPT-4o (LLM Evaluator)	79.0%	85.2%	79.0%	78.0%
	AWS Bedrock	80.9%	85.5%	80.9%	80.3%

Table 2: Evaluation results across six datasets comparing cross-encoder, LLM-based, and AWS Bedrock hallucination detection methods. Results are reported as Accuracy, Macro Precision, Macro Recall, and Macro F1. RetailbankQA shows the largest gains from the cross-encoder approach.

3 Model

The model used for binary hallucination classification is a fine-tuned cross-encoder based on the **DeBERTa-v3-base** architecture.[7] This transformer has demonstrated performance on natural language inference (NLI) benchmarks such as MNLI, ANLI, and SNLI, outperforming models like BERT, RoBERTa, and XLNet (He et al., 2021; Liu et al., 2019; Yang et al., 2019).

DeBERTa-v3-base is well-suited for hallucination detection tasks that require fine-grained factual consistency checks, due to its pretraining on entailment-style objectives-that is, tasks where the model learns to assess whether one sentence is logically supported by another. Compared to bi-encoders or prompt-based LLM evaluators, the cross-encoder architecture jointly encodes the *query*, *retrieved context*, and *response*, enabling the model to capture rich inter-component dependencies. This design is consistent with recent hallucination detection frameworks such as Luna (Belyi et al., 2024) and Lettuce Detect (Ádám Kovács and Recski, 2025), and aligns with answer faithfulness metrics used by LLM evaluation protocols.

3.1 Training Setup

To adapt the pretrained DeBERTa-v3-base model, a classification head is added to produce a single logit representing the probability that the response is grounded. Inputs are tokenized in the format:

```
[CLS] query [SEP] context [SEP] response [SEP]
```

Training is performed using the Hugging Face `Trainer` API based on three epochs, a batch size of 8, a learning rate of 1×10^{-5}, and the `AdamW` optimizer (weight decay = 0.01). This configuration was selected based on a targeted grid search, where it achieved the most stable validation performance and highest F1 score, while reducing overfitting and training variance associated with smaller batch sizes.[8] Binary cross-entropy loss is applied, with labels encoded as 1 for grounded (`PASS`) and 0 for hallucinated (`FAIL`). The training dataset is balanced and composed of examples from both HaluBench (general domain) and RetailBankQA (financial domain). The Hugging Face DeBERTa tokenizer is used to tokenize all inputs.

3.2 Context Window Constraints

Context length limitations present a challenge for groundedness assessment of documents. To address the 512-token constraint of DeBERTa-v3-base, a windowed chunking strategy is applied: each input is split into overlapping context chunks and passed independently through the model. Groundedness probabilities are computed for each chunk, and the maximum score across

[7] https://huggingface.co/microsoft/deberta-v3-base

[8] Hyperparameter search results available at GitHub.

chunks is used to derive a single, example-level binary classification.

Token length distribution analysis (see Figure 2) indicates that around 94% of test cases are within the 512-token limit. Given this distribution, more computationally heavy token-level span attribution was considered, but not implemented, particularly as longer contexts were largely to one subdataset which already had known limitations of containing numeric and calculation-heavy test cases (FinanceBench, where 44.3% of examples exceed the threshold, compared to 2.5% in DROP, 1.5% in PubMedQA, and 0.67% in RAGTruth).

This design aligns with Luna (Belyi et al., 2024), which also adopts a windowed chunking strategy for processing long RAG inputs. However, unlike Luna—which performs token-level span classification to identify hallucinated spans—this implementation focuses on example-level prediction to reduce computational overhead. As language models evolve to support longer context windows and multi-turn dialogue histories, token-level grounding and span attribution remain important directions for future work.

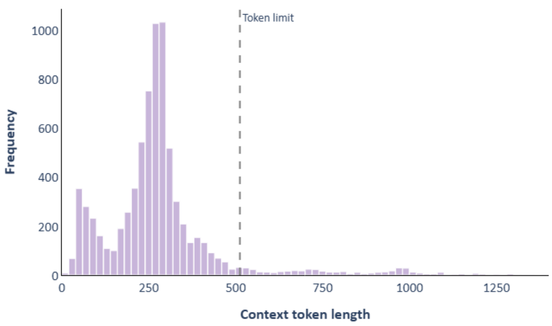

Figure 2: Context Token Length Distribution for full dataset (dashed line = 512-token max input for DeBERTa-v3-base).

4 Baseline Comparison Models

Baseline comparisons include a prompted GPT-4o evaluator and a commercial grounding guardrail from AWS Bedrock.

4.1 Prompted LLM Verifier

A zero-shot evaluation baseline is implemented using OpenAI's `gpt-4o-2024-05-13` model, accessed via the OpenAI API.[9] The prompting approach is adapted from the Lynx framework proposed by Ravi et al. (2024) and is presented using a structured Jinja2 template.The full prompt is included in Appendix B. Each input is formatted as a (query, document, response) triplet, and the model is instructed to assess whether the response is grounded in the retrieved document, based on strict faithfulness criteria.

The model returns a JSON object containing a `REASONING` field (a brief justification) and a `SCORE` field, which is parsed as either `"PASS"` or `"FAIL"`. These binary outputs are used for groundedness classification in the evaluation.

4.2 AWS Bedrock Guardrail

As an off-the-shelf comparator model, the commercial contextual grounding guardrail from Amazon Bedrock was selected as a closed-source baseline.[10] Each (query, context, response) triplet is submitted via the `apply_guardrail` API with contextual grounding policies enabled and relevance checks disabled. The system uses Claude Sonnet 4, the foundation model available via AWS at the time of evaluation, to return a grounding score and a threshold-based decision.

A fixed threshold of 0.7 is applied to determine whether the response is grounded. Outputs are mapped to binary `PASS` or `FAIL` labels to support direct comparison with other evaluators. Although the guardrail is configured to block ungrounded responses, its decision is used solely for classification in this study. This baseline was selected for its alignment with the contextual grounding task and its accessibility as a managed service.

4.3 Evaluation

Model evaluation was conducted on two test sets: the HaluBench hold-out set and the synthetic RetailbankQA test set. The classifier produces chunk-level predictions, which are then aggregated to the example level using the maximum predicted probability across all chunks for each test case.

Performance is reported using example-level accuracy, precision, recall, and F1-score, with `PASS` indicating a grounded response and `FAIL` indicating a hallucinated one. Evaluation includes classification reports for each source dataset and confusion matrix breakdowns. Results are further stratified by source domain (e.g., FinanceBench, DROP) to

[9] https://platform.openai.com/docs/models/gpt-4o

[10] https://docs.aws.amazon.com/bedrock/latest/userguide/guardrails-contextual-grounding-check.html

assess generalization under domain shift. Qualitative failure cases are also reviewed to understand where the model struggles with subtle hallucination patterns or ambiguous references.

5 Results and discussion

Results in Figure 3 show a comparative evaluation of the cross-encoder classifier and two LLM-based evaluators. Metrics include example-level accuracy, macro precision, macro recall, and macro F1.

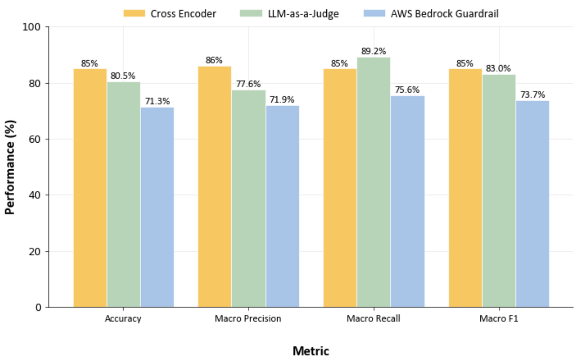

Figure 3: Performance comparison of cross-encoder versus LLM-based models; by evaluation metric.

The cross-encoder achieves consistent scores across all metrics, with macro precision and recall of 88.5% and 88.0%, respectively. Its macro F1 score (88.2%) is slightly above that of the GPT-4o-based evaluator (85.7%) and notably higher than the AWS Bedrock guardrail (79.7%). These results reflect weighted averages across the held-out evaluation sets.

The cross-encoder's deterministic outputs and lower latency suggest it may be a practical option for integration into production systems, particularly in domains requiring bounded inference costs and reproducibility.

That said, the performance gap between the cross-encoder and the GPT-4o evaluator is modest. LLM-based methods remain competitive—particularly in low-throughput settings where qualitative reasoning or explanatory outputs are valuabl

5.1 Error Analysis

[Discuss types of hallucinations missed, false positives, patterns across datasets.]

6 Conclusion

[Summary of findings. Implications for scalable, reliable financial dialogue systems. Future work.]

Limitations

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A RetailbankQA Triplet with Grounded and Hallucinated Responses

QUERY *How Long Will My Transaction Take?*
CONTEXT When making a transfer, the time it takes to reach the recipient can vary depending on the type of transfer. Faster Payments usually arrive within minutes, while BACS payments can take up to three working days. International transfers may take longer depending on the destination and receiving bank.
RESPONSE (Grounded) The timing for your transaction to be completed depends on the transfer type. Faster Payments typically arrive within minutes, BACS transfers can take up to three working days, and international transfers may take longer depending on the recipient's bank.
RESPONSE (Hallucinated) When you send a transfer, it will always arrive instantly, regardless of the method or country. All transactions are processed in real time by default, including international ones.
ANNOTATION **Span:** "it will always arrive instantly" and "including international ones" **Reason:** Original: "Faster Payments usually arrive within minutes… BACS… up to three working days… international transfers may take longer" Generative: "all transactions are processed in real time… including international ones"

Table 3: Example of a Query-Conext-Response triplet from synthetic RetailbankQA dataset with grounded and hallucinated response.

B Groundedness Evaluation Prompt

The following prompt was used for LLM-based groundedness evaluation (GPT-4o). Query, context, and response fields were dynamically filled for each example.

```
Given the following QUESTION, DOCUMENT and ANSWER
you must analyze the provided answer and
determine whether it is faithful to the contents
of the DOCUMENT.
A faithful ANSWER must not offer new information
beyond the context provided in the DOCUMENT.
A faithful ANSWER must be grounded in the
DOCUMENT and not based on external knowledge.
A faithful ANSWER also must not contradict
information provided in the DOCUMENT.
Even if there is a slight discrepancy between the
ANSWER and the DOCUMENT, or any ungrounded
assumptions made in the ANSWER, the ANSWER must
be considered not faithful.
The faithfulness of the ANSWER is independent of
its relevancy to QUESTION.
Even if the ANSWER is irrelevant to the QUESTION,
your task is to evaluate if it is faithful to the
DOCUMENT regardless of relevancy.
Output your final verdict by strictly following
this format: "PASS" if the answer is faithful to
the DOCUMENT and "FAIL" if the answer is not
faithful to the DOCUMENT. Briefly, show your
reasoning.

–
QUESTION (THIS DOES NOT COUNT AS BACKGROUND
INFORMATION):
{{user_query}}
–
DOCUMENT:
{{retrieved_context}}
–
ANSWER:
{{model_answer}}
–
Your output should be in JSON format with the
keys "REASONING" and "SCORE":
{"REASONING": <brief reasoning as bullet points>,
"SCORE": <your final score>}
```